

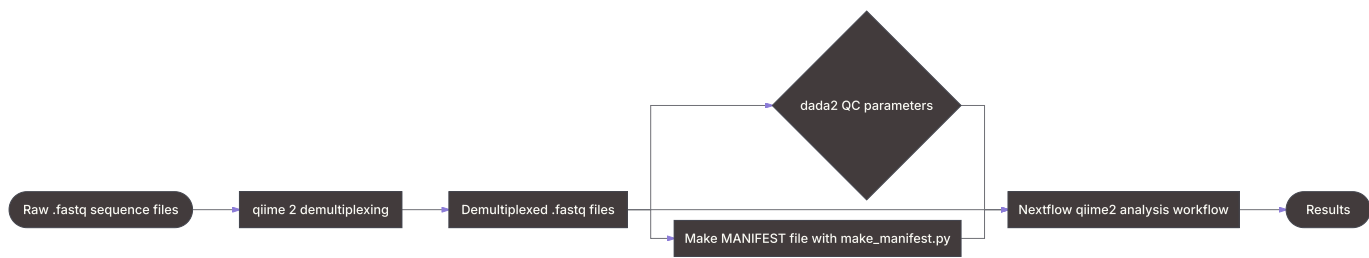
Qiime2 16S paired-end demultiplexed sequence analysis pipeline with Nextflow

Author: Patricia Zito

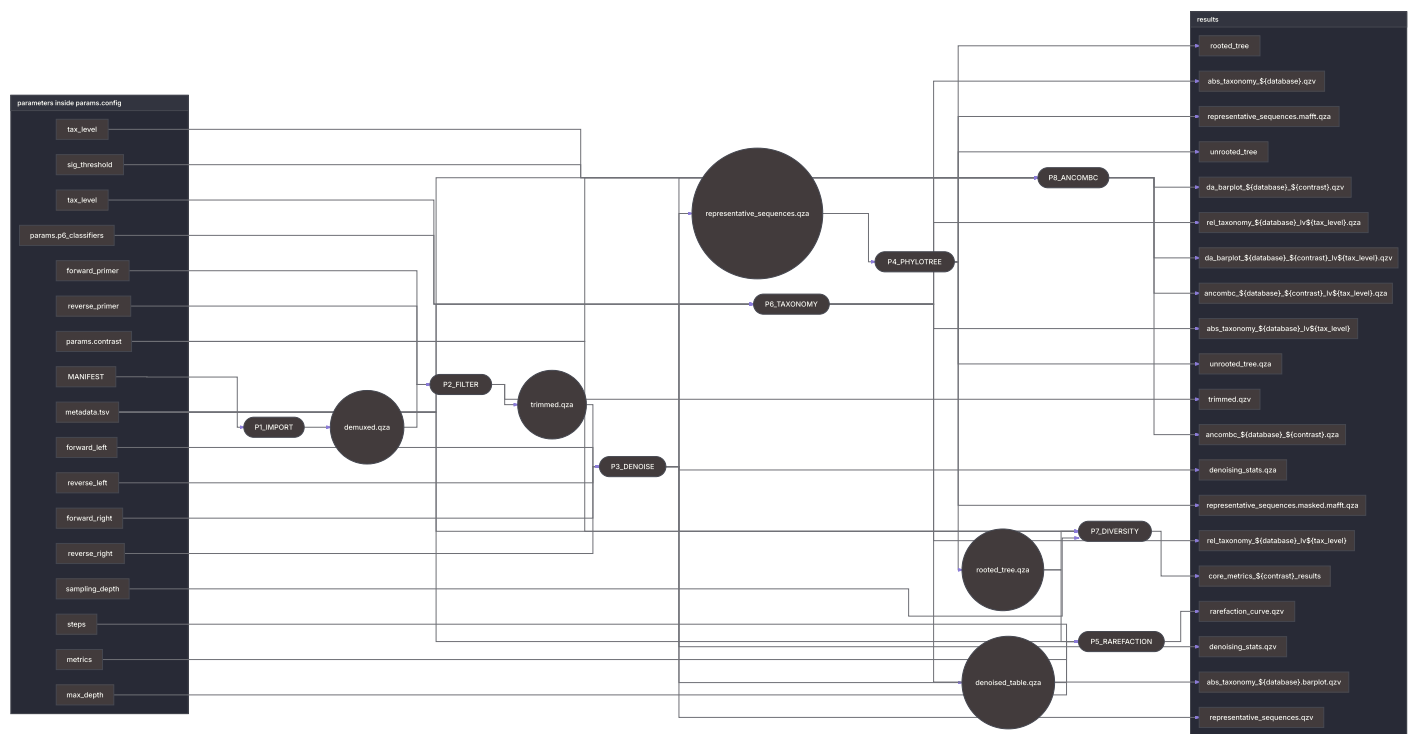
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Successfully ran in lab server machine.

Whole Analysis Overview



Nextflow Workflow Overview



Project Folder Structure:

```
(base) leelab@IBIO-JH5DN34:/media/leelab/HDD_Array/PZito/16s/2013_04_17-  
nf$ tree .
```

```
.  
├── demultiplexed_seqs  
│   ├── 20130417_LisleVerte_copepod_6_L001_R1_001.fastq.gz  
│   ├── 20130417_LisleVerte_copepod_6_L001_R2_001.fastq.gz  
│   ├── 20130417_LisleVerte_water_4_L001_R1_001.fastq.gz  
│   ├── 20130417_LisleVerte_water_4_L001_R2_001.fastq.gz  
│   ├── 20130417_Racine_copepod_1_L001_R1_001.fastq.gz  
│   ├── 20130417_Racine_copepod_1_L001_R2_001.fastq.gz  
│   ├── 20130417_Racine_water1_5_L001_R1_001.fastq.gz  
│   ├── 20130417_Racine_water1_5_L001_R2_001.fastq.gz  
│   ├── 20130417_Racine_water2_2_L001_R1_001.fastq.gz  
│   ├── 20130417_Racine_water2_2_L001_R2_001.fastq.gz  
│   ├── 20130417_Racine_water3_3_L001_R1_001.fastq.gz  
│   ├── 20130417_Racine_water3_3_L001_R2_001.fastq.gz  
│   └── MANIFEST  
├── metadata_20130417.tsv  
├── results  
│   ├── abs_taxonomy_silva-full.barplot.qzv  
│   ├── abs_taxonomy_silva-full_lv6  
│   │   ├── abs_taxonomy_silva-full_lv6.tsv  
│   │   └── feature-table.biom  
│   ├── abs_taxonomy_silva-full_lv6.qza  
│   ├── abs_taxonomy_silva-full.qza  
│   ├── abs_taxonomy_silva-full.qzv  
│   ├── ancombc_silva-full_host_adapted_salinity_lv6.qza  
│   ├── ancombc_silva-full_host_adapted_salinity.qza  
│   ├── ancombc_silva-full_sample_type_lv6.qza  
│   ├── ancombc_silva-full_sample_type.qza  
│   ├── core_metrics_host_adapted_salinity_results  
│   │   ├── bray_curtis_distance_matrix.qza  
│   │   ├── bray_curtis_emperor.qzv  
│   │   ├── bray_curtis_pcoa_results.qza  
│   │   ├── evenness_significance.qzv  
│   │   ├── evenness_vector.qza  
│   │   ├── faith_pd_significance.qzv  
│   │   ├── faith_pd_vector.qza  
│   │   ├── jaccard_distance_matrix.qza  
│   │   ├── jaccard_emperor.qzv  
│   │   ├── jaccard_pcoa_results.qza  
│   │   ├── observed_features_vector.qza  
│   │   ├── rarefied_table.qza  
│   │   ├── shannon_vector.qza  
│   │   ├── unweighted_unifrac_distance_matrix.qza  
│   │   └── unweighted_unifrac_emperor.qzv
```

```
| | | └─ unweighted_unifrac_host_adapted_salinity.qzv
| | | └─ unweighted_unifrac_pcoa_results.qza
| | | └─ weighted_unifrac_distance_matrix.qza
| | | └─ weighted_unifrac_emperor.qzv
| | | └─ weighted_unifrac_pcoa_results.qza
| └─ core_metrics_sample_type_results
| | | └─ bray_curtis_distance_matrix.qza
| | | └─ bray_curtis_emperor.qzv
| | | └─ bray_curtis_pcoa_results.qza
| | | └─ evenness_significance.qzv
| | | └─ evenness_vector.qza
| | | └─ faith_pd_significance.qzv
| | | └─ faith_pd_vector.qza
| | | └─ jaccard_distance_matrix.qza
| | | └─ jaccard_emperor.qzv
| | | └─ jaccard_pcoa_results.qza
| | | └─ observed_features_vector.qza
| | | └─ rarefied_table.qza
| | | └─ shannon_vector.qza
| | | └─ unweighted_unifrac_distance_matrix.qza
| | | └─ unweighted_unifrac_emperor.qzv
| | | └─ unweighted_unifrac_pcoa_results.qza
| | | └─ unweighted_unifrac_sample_type.qzv
| | | └─ weighted_unifrac_distance_matrix.qza
| | | └─ weighted_unifrac_emperor.qzv
| | | └─ weighted_unifrac_pcoa_results.qza
| └─ da_barplot_silva-full_host_adapted_salinity_lv6.qzv
| └─ da_barplot_silva-full_host_adapted_salinity.qzv
| └─ da_barplot_silva-full_sample_type_lv6.qzv
| └─ da_barplot_silva-full_sample_type.qzv
| └─ demuxed.qza
| └─ denoised_table.qza
| └─ denoising_stats.qza
| └─ denoising_stats.qzv
| └─ rarefaction_curve.qzv
| └─ rel_taxonomy_silva-full_lv6
| | | └─ feature-table.biom
| | | └─ rel_taxonomy_silva-full
| └─ rel_taxonomy_silva-full_lv6.qza
| └─ representative_sequences.mafft.qza
| └─ representative_sequences.masked.mafft.qza
| └─ representative_sequences.qza
| └─ representative_sequences.qzv
| └─ rooted_tree
| | | └─ tree.nwk
| └─ rooted_tree.qza
```

```
|   ├── trimmed.qza
|   ├── trimmed.qzv
|   ├── unrooted_tree
|   |   └── tree.nwk
|   └── unrooted_tree.qza
└── scripts
    ├── main_test.nf
    ├── make_manifest.py
    ├── nextflow.config
    ├── P1_import.nf
    ├── P2_filter.nf
    ├── P3_denoise.nf
    ├── P4_phylotree.nf
    ├── P5_rarefaction.nf
    ├── P6_taxonomy.nf
    ├── P7_diversity.nf
    ├── P8_ancombc.nf
    └── params.config
└── work
```

Usage Example:

```
# importing raw fastq files
qiime tools import --type EMPPairedEndSequences --input-path
raw_reads_2013_04_17/reads/ --output-path
raw_reads_2013_04_17/raw_reads_2013_04_17.qza

# demultiplexing
qiime demux emp-paired --m-barcodes-file
raw_reads_2013_04_17/metadata_20130417.tsv --m-barcodes-column barcode --
i-seqs raw_reads_2013_04_17/raw_reads_2013_04_17.qza --o-per-sample-
sequences demultiplexed/demultiplexed_2013_04_17 --o-error-correction-
details demultiplexed/demultiplexing_details.qza --verbose

# make manifest file
python3 scripts/make_manifest.py -m metadata.tsv -s demultiplexed_seqs -o
demultiplexed_seqs/MANIFEST

# run the pipeline
nextflow scripts/main.nf --with-report -profile lab_server

# tarball it
tar -cvf 2013_04_17-nf.results.tar.gz results/
```